

PHARMACY AND THE ENVIRONMENT

FYI

- Do you think about the environmental impact when you are deciding which drug to suggest to a patient or healthcare provider?
- This website allows you to better understand the environmental impact a drug has. It discusses persistence, bioaccumulation and toxicity



Janusinfo REGION STOCKHOLM | Pharmaceuticals and Environment

- Impact of inhalers on the atmosphere

A top-down view of a grey desk with various medical and study items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop.

Pharm*Achieve*

SOAP NOTES

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

Providing Care:
Clinical Care

Providing Care:
Distribution

Knowledge
& Expertise

Communication
& Collaboration

Leadership &
Stewardship

Professionalism

MEDICAL DOCUMENTATION

- Medical documentation is a crucial aspect of providing excellent patient care
- Accurate documentation can improve patient care, reduce errors and promote interprofessional collaboration
- A **SOAP (Subjective, Objective, Assessment, Plan)** note is a structured approach to documentation that ensures that **key information is clearly communicated between healthcare providers**
- A **DAP (Data, Assessment, Plan)** note is another commonly used term for SOAP
- It is widely accepted as the standard and most appropriate way to document medical findings
- It helps team members evaluate information and understand how a diagnosis is made

SOAP NOTES

In a DAP note, the **D** = **S** + **O**

S	Subjective	Chief complaint, history of present illness, SCHOLARHAMS acronym (symptoms, characteristics, history, onset, location, aggravating factors, remitting factors, health conditions, allergies, medications, social history), patient's goals and preferences
O	Objective	Vital signs, labs, imaging, physical exam results, other diagnostic test results
A	Assessment	Based on the gathered data, what are the drug therapy problems? what is the most likely diagnosis? What are the differential diagnoses? What are the therapeutic goals/targets?
P	Plan	How will it be managed? Are additional tests needed? Does the patient need to be referred to a specialist? What medications should the patient be on? Does the patient require education or counselling? What is the monitoring and follow-up plan?

EXAMPLE

SL is a 65-year-old male who has been recently admitted overnight to the emergency department. He has a history of COPD and is currently taking tiotropium 18 mcg daily and salbutamol 100 mcg 2 puffs QID PRN at home. He is experiencing dyspnea and increased sputum production (yellow/green in colour).

Examination: BP = 130/73 mmHg; HR = 115 bpm; temperature = 38.1 degrees Celsius; WBC = 15.1; Neutrophils = 11.6; Oxygen saturation = 80%; pH = 7.27. Blood and sputum cultures are drawn and sent for sensitivity and culture. On his previous admission to the hospital 2 months ago, the patient's sputum culture grew *Pseudomonas aeruginosa*. He is provisionally diagnosed with mixed hypoxic and hypercapnic respiratory failure secondary to acute exacerbation of COPD and pneumonia.

Based on the patient's clinical presentation and past medical history, the physician decides to start SL on cefepime 2 g Q8H and then will switch to targeted therapy once cultures return.

Categorize the information above into the appropriate headings.

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Categorize the information above into the appropriate headings.

Subjective	65 yo male, history of COPD, admitted overnight, takes tiotropium 18 mcg daily and salbutamol 100 mcg 2 puffs QID PRN
Objective	BP = 130/73 mmHg; HR = 115 bpm; temperature = 38.1 degrees Celsius; WBC = 15.1; Neutrophils = 11.6; Oxygen saturation = 80%; pH = 7.27
Assessment	Mixed hypoxic and hypercapnic respiratory failure secondary to acute exacerbation of COPD and pneumonia
Plan	Cefepime 2 g Q8H followed by targeted once blood and sputum cultures return

CASE #1

You are a pharmacist working at a 50-bed nursing home in Vancouver. The doctor in charge visits once a week and follows up on any concerns that staff bring to his attention in the doctor's book.

This week, you are particularly concerned about an 88-year-old woman, JJ, who, as nurses state, has been refusing her anti-hypertensive medications recently. JJ stays in Bed 101 at the nursing home, as she has a nice patio that she can sit out in. JJ states that she takes too many medications and wants to reduce the number of pills she takes per day. She currently takes donepezil 10 mg PO daily for her dementia, ramipril 5 mg PO daily, furosemide 20 mg PO daily, and metoprolol 50 mg PO BID for her hypertension. You want her to start taking a vitamin D supplement too, but you know that she is not going to want to take more medication. Given that she often does not take her medications, her blood pressure has been high, averaging 170/110 mmHg this week. Nurses state that she has not been displaying any signs or symptoms of hypertension or distress. She is oriented to time and place when you speak with her. She is eating and sleeping normally. You decide to ask the nurses to monitor her blood pressure twice a day and record it in her notes. You also think that the doctor should consider modifying her anti-hypertensive regimen to reduce her pill burden and improve adherence. If this works, you can speak to the patient about adding a Vitamin D supplement at your next visit in 2 weeks. You know she has no known drug allergies (NKDA).

Write a SOAP note in the doctor's book for the doctor to read and follow up on when he visits the nursing home tomorrow.

CASE #1 SOAP NOTE

JJ is an 88-year-old female (Bed 101) who has been refusing to take her anti-hypertensives most days this week. She is asymptomatic, eating and sleeping normally and is not displaying any signs of distress. JJ expressed that she would like to reduce her pill burden. Diagnoses include hypertension and dementia. Medication regimen consists of donepezil 10 mg PO daily, ramipril 5 mg PO daily, furosemide 20 mg PO daily and metoprolol 50 mg PO BID. She has NKDA and does not take any medications for bone protection.

Average blood pressure this week: 170/110 mmHg. Oriented to time and place.

Patient appears to be experiencing elevated blood pressure due to non-adherence to medication regimen (currently prescribed 3 anti-hypertensives, including one that is dosed BID). Patient is also at risk of Vitamin D deficiency.

Doctor to kindly consider discontinuing metoprolol 50 mg BID first, to eliminate one medication administration time (evening) and reduce pill burden. Nurses to monitor and note blood pressure twice daily to assess impact of new regimen. Doctor to consider starting Vitamin D 1000 units PO daily for bone protection once blood pressure stable and if patient agrees.

CASE #2

You are a pharmacist working at a busy downtown Calgary hospital in the endocrinology unit. This is your first year as a hospital pharmacist and you are keen to learn as much as you can on the unit.

You are managing a patient, RT, who is a 65-year-old female with type 2 diabetes and dyslipidemia. Her HbA1C target is $< 6.5\%$ and her BMI is 28 kg/m^2 . Despite being adherent to her medications (metformin 1000 mg PO BID, gliclazide immediate-release (IR) 80 mg PO daily, ramipril 10 mg PO daily, and atorvastatin 40 mg PO daily), RT's glycemic control remains suboptimal after 4 months of therapy. Her fasting blood glucose has been between 13.2-15.3 mmol/L for the past 2 weeks. She came to the hospital because she has been feeling dizzy, weak and nauseous for the past 4 days. She enjoys smoking a pack a day but finds that recently she has been feeling more nauseous. You decide you want to speak to her about quitting before she is discharged. She has tried to sleep more and take prochlorperazine, but nothing has helped. Her blood pressure has consistently been $<130/80 \text{ mmHg}$ and her creatinine clearance (CrCl) is 75 mL/min. Her HbA1C is 8.5%. Her LDL is 3.5 mmol/L.

You are concerned about the potential long-term cardiovascular implications. You think RT should start a third anti-hyperglycemic agent with proven cardiovascular benefits. After much consideration, you decide to suggest empagliflozin 10 mg PO daily.

CASE #2 SOAP NOTE

RT is a 65-year-old female who presented to the hospital with dizziness, nausea and weakness. She tried to take prochlorperazine for her nausea, but it did not help, and smoking worsens it. Her medical history is significant for diabetes and dyslipidemia. Medications include metformin 1000 mg PO BID, gliclazide IR 80 mg PO daily, ramipril 10 mg PO daily, and atorvastatin 40 mg PO daily. She smokes a pack of cigarettes a day.

Patient's HbA1C is 8.5% (target 6.5%), LDL 3.5 mmol/L, and CrCl 75 mL/min. Fasting blood glucose this week ranged from 13.2-15.3 mmol/L and BP has been <130/80 mmHg throughout her admission. BMI is 28 kg/m².

Patient is experiencing uncontrolled diabetes and symptomatic hyperglycemia. She would benefit from an additional medication to improve diabetes control and weight management and reduce the risk of major adverse cardiac effects. Patient would also benefit from discussion on smoking cessation.

Doctor to kindly consider addition of empagliflozin 10 mg PO daily. Pharmacy team will speak to patient about smoking cessation prior to discharge and will update medical team on plan. Nurses to monitor fasting blood glucose daily until discharge.

REFERENCES

1. Podder V, Lew V, Ghassemzadeh S. SOAP Notes. 2024. StatPearls. <https://www.ncbi.nlm.nih.gov/books/NBK482263/#:~:text=The%20Subjective%2C%20Objective%2C%20Assessment%20and,%5D%5B2%5D%5B3%5D>
2. Sindhu K. 2020. What are SOAP notes? Wolters Kluwer. <https://www.wolterskluwer.com/en/expert-insights/what-are-soap-notes>.
3. Breault R, Barnes D, Moulton D, Salokangas E. *Guide to Pharmacist Documentation of Patient Care: A Companion Guide to the Patient Care Process*. Faculty of Pharmacy + Pharmaceutical Sciences, University of Alberta; 2025. https://www.ualberta.ca/en/pharmacy/media-library/guide-to-pharmacist-documentation-of-patient-care_fopps2025-3.pdf

CHANGE LOG

July 2025

- Formatting and grammar changes throughout
- Slide 8: Modified allergy statement to include NKDA abbreviation
- Slide 9: Changed IU to units
- Slide 10: Changed initial dose of empagliflozin, changed eGFR to CrCl, added 'immediate-release (IR)' to gliclazide
- Slide 11: Changed eGFR to CrCl, removed repeat mention of CrCl in the second paragraph, added IR to gliclazide

October 2025

- Slide 8: Minor formatting and phrasing edits made

March 2026

- Slide 3: Updated NAPRA competencies
- Slides 4 & 5: Added information on DAP notes
- Slide 5: Updated with more information
- Slide 12: Updated references